

A Reinforcement Learning Approach for Continuum Robot Control

连续体机器人控制的强化学习方法

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Abstract

摘要

Rigid joint manipulators are limited in their movement and degrees of freedom (DOF), while continuum robots possess a continuous backbone that allows for free movement and multiple DOF. Continuum robots move by bending over a section, taking inspiration from biological manipulators such as tentacles and trunks. This paper presents an implementation of a forward kinematics and velocity kinematics model to describe the planar continuum robot, along with the application of reinforcement learning (RL) as a control algorithm. In this paper, we have adopted the planar constant curvature representation for the forward kinematic modeling. This choice was made due to its straightforward implementation and its potential to fill the literature gap in the field RL-based control for planar continuum robots. The intended control mechanism is achieved through the use of Deep Deterministic Policy Gradient (DDPG), a RL algorithm that is suited for learning controls in continuous action spaces. After simulating the algorithm, it was observed that the planar continuum robot can autonomously move from any initial point to any desired goal point within the task space of the robot. By analyzing the results, we wanted to recommend a future direction for research in the field of continuum robot control, specifically in the application of RL algorithms. One potential area of focus could be the integration of sensory feedback, such as vision or force sensing, to improve the robot's ability to navigate complex environments. Additionally, exploring the use of different RL algorithms, such as Proximal Policy Optimization (PPO) or Trust Region Policy Optimization (TRPO), could lead to further advancements in the field. Overall, this paper demonstrates the potential for RL-based control of continuum robots and highlights the importance of continued research in this area.

刚性关节机械臂在运动和自由度 (DOF) 方面存在限制, 而连续体机器人拥有连续的骨架, 允许自由运动和多自由度。连续体机器人通过在某一段弯曲来移动, 其灵感来源于生物机械臂, 如触手和象鼻。本文提出了用于描述平面连续体机器人的正向运动学和速度运动学模型的实现, 并应用强化学习 (RL) 作为控制算法。本文采用了平面恒曲率表示法进行正向运动学建模。选择该方法是因为其实现简便, 并有望填补基于强化学习的平面连续体机器人控制领域的文献空白。所采用的控制机制是通过深度确定性策略梯度 (DDPG) 算法实现的, 该强化学习算法适合于连续动作空间的控制学习。通过算法仿真观察到, 平面连续体机器人能够自主地从任意初始点移动到机器人任务空间内的任意目标点。通过结果分析, 我们希望为连续体机器人控制领域的未来研究方向提出建议, 特别是在强化学习算法的应用方面。一个潜在的研究方向是整合感知反馈, 如视觉或力觉传感, 以提升机器人在复杂环境中的导航能力。此外, 探索不同的强化学习算法, 如近端策略优化 (PPO) 或信赖域策略优化 (TRPO), 可能推动该领域的进一步发展。总体而言, 本文展示了基于强化学习的连续体机器人控制的潜力, 并强调了该领域持续研究的重要性。

Highlights

亮点

- Implementing the forward kinematics and velocity kinematics model for the planar case of a continuum robot
- 实现了平面连续体机器人的正向运动学和速度运动学模型
- Reinforcement learning has been applied as a control algorithm for continuous robot control
- 将强化学习应用于连续机器人控制的控制算法
- Validating results obtained by applying the Deep Deterministic Policy Gradient (DDPG) reinforcement learning algorithm
- 验证了应用深度确定性策略梯度 (DDPG) 强化学习算法获得的结果

1 Introduction

1 引言

The concept behind continuum robots is inspired by the natural abilities of certain biological organisms, such as snakes, elephant trunks, octopus tentacles, and similar creatures [1]. Unlike traditional rigid-joint manipulators, continuum robots bend individually for each section by creating a sequence of continuous non-joint arcs [2]. After extensive research and development in academic settings, these designs have recently been applied to large-scale grasping, movement, and positioning in industrial contexts, as well as search and rescue operations in confined spaces [8]. They have also gained popularity in surgical applications [9], as they offer adaptable qualities that are particularly advantageous in medical contexts. With the ability to manipulate objects with an infinite number of degrees of freedom (DOF) on small scales, continuum robots provide flexible access for the target and less intrusion for the patient.

连续体机器人的概念灵感来源于某些生物体的自然能力，如蛇、象鼻、章鱼触手等 [1]。与传统的刚性关节机械臂不同，连续体机器人通过形成一系列连续的非关节弧段，实现每个部分的独立弯曲 [2]。经过学术界的广泛研究和开发，这些设计最近已应用于工业领域的大规模抓取、移动和定位，以及狭小空间的搜救行动 [8]。它们在外科手术应用中也越来越受欢迎 [9]，因为其适应性特性在医疗环境中尤为有利。连续体机器人能够在小尺度上以无限自由度 (DOF) 操作物体，提供灵活的目标接触方式，同时对患者的侵入性较小。

Conventional rigid joint manipulators, also known as discrete manipulators, have a limited number of operable, series-independent, rigid connections [2]. Despite their limitations, they are highly practical for many tasks.

However, in constrained environments, they may not be able to reach the desired end-effector position, resulting in failure to meet location criteria. Another disadvantage of discrete manipulators is that they require a specific tool, such as a gripper or hand, to manipulate an object. In some cases, it may be preferable to use a manipulator in sections, such as the elephant trunk, to hold objects. Adding more joints to discrete manipulators increases their flexibility and maneuverability, resulting in redundancy and maneuverability. This elevates them to the second category of manipulators, known as serpentine robots, which includes hyper-redundant manipulators. Continuum robots, on the other hand, lack rigid joints and bend continuously along their length, similar to biological bodies and tentacles. This makes continuum robots more maneuverable than discrete and serpentine manipulators, but also makes their kinematics more complex. See Fig. 1.

传统的刚性关节机械臂，也称为离散机械臂，具有有限数量的可操作、串联独立的刚性连接 [2]。尽管存在局限性，它们在许多任务中仍然非常实用。然而，在受限环境中，它们可能无法达到期望的末端执行器位置，导致无法满足定位要求。离散机械臂的另一个缺点是需要特定工具，如夹持器或手部，来操作物体。在某些情况下，使用分段机械臂 (如象鼻) 来抓持物体可能更为合适。增加离散机械臂的关节数量可以提升其灵活性和机动性，产生冗余度和机动性，从而使其升格为第二类机械臂，即蛇形机器人，包括超冗余机械臂。相比之下，连续体机器人没有刚性关节，沿长度连续弯曲，类似于生物体和触手。这使得连续体机器人比离散和蛇形机械臂更具机动性，但其运动学也更为复杂。见图 1。

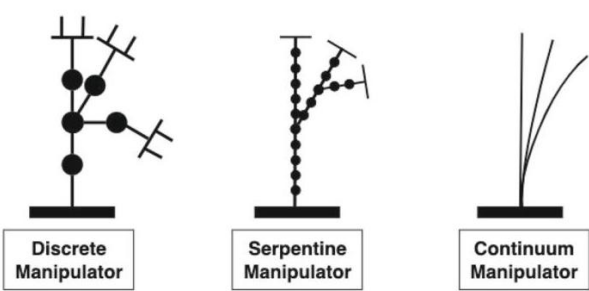


Fig. 1 Class of manipulators[2]

图 1 机械臂类别 [2]

There are numerous methods for controlling continuum robots, such as PID and predictive control, which are

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based on explicitly implemented kinematics or dynamics [3-6]. Continuum robots, with their inherent flexibility and high degree of freedom, present unique control challenges that are not adequately addressed by existing techniques. As highlighted by [7], the soft and continuously deformable nature of continuum robots introduces significant complexities in their control. Traditional control methods often rely on precise mathematical models and struggle to handle the nonlinear dynamics and uncertainties associated with continuum robots. Furthermore, these methods are typically designed for rigid robots and do not fully exploit the potential of soft robots' embodied intelligence.

控制连续体机器人有多种方法,如基于明确实现的运动学或动力学的PID和预测控制[3-6]。连续体机器人因其固有的柔性和高度自由度,带来了独特的控制挑战,而现有技术尚未充分解决这些问题。如文献[7]所指出,连续体机器人软性且连续可变形的特性使其控制复杂性显著增加。传统控制方法通常依赖精确的数学模型,难以处理连续体机器人所涉及的非线性动力学和不确定性。此外,这些方法通常为刚性机器人设计,未能充分利用软体机器人所具备的体现智能(embodied intelligence)潜力。

In contrast, Reinforcement Learning (RL) offers a promising alternative for controlling continuum robots. RL algorithms learn from interactions with the environment and adapt their control policies accordingly, making them well-suited to handle the complexities and uncertainties of continuum robots. Moreover, RL does not require a precise model of the robot and can learn effective control policies even in the presence of model inaccuracies. Therefore, we believe that RL holds significant potential for advancing the control of continuum robots and addressing the limitations of existing techniques.

相比之下,强化学习(Reinforcement Learning, RL)为控制连续体机器人提供了有前景的替代方案。RL算法通过与环境的交互学习并相应调整控制策略,使其非常适合处理连续体机器人的复杂性和不确定性。此外,RL不依赖于机器人精确模型,即使存在模型误差,也能学习到有效的控制策略。因此,我们认为RL在推动连续体机器人控制发展和克服现有技术局限方面具有重要潜力。

RL is a cost-effective approach for controlling robots or solving complex control problems. It involves training a learnable architecture based on generated reward or penalty values through trial and error. RL has become increasingly popular due to its ability to handle control problems, make challenging sequential decisions, and solve a wide range of issues, even before being combined with deep learning. However, combining deep learning techniques with RL has increased the number of problems that have been solved, as well as their success rate [10]. This combination works better for high-dimensional state and action spaces. Deep reinforcement learning (Deep RL) has been used in many applications to surpass human levels. There have been several significant experiments using deep RL in games, including beating even the best poker players in the world [11, 12], playing video games from pixels [13, 14], and mastering the game of Go with programs like AlphaGo [15]. RL and Deep RL are used in a variety of academic and commercial contexts, including robotics [16-19], autonomous vehicles [20], banking [21], smart grids [22], and the energy sector [23]. RL in robotics allows robots to interact with their environment and learn the optimal path or behavior through trial and error. Instead of explicitly defining the mathematical details of a problem's solution, the control task designer in RL provides feedback in the form of a scalar objective function that measures the robot's performance over one step. There exist multiple RL techniques that can be employed to address the control problem in continuum robots [24-27]. For example, [26] employed a control strategy grounded in deep Q learning to successfully accomplish 2D position reaching. Additionally, [27] proposed a control method that does not rely on a specific model, utilizing Q learning. This method was then applied to a multi-segment soft manipulator in order to achieve 2D position reaching.

强化学习是一种成本效益高的机器人控制或复杂控制问题求解方法。它通过试错过程，根据生成的奖励或惩罚值训练可学习的架构。由于其处理控制问题、做出复杂序列决策及解决广泛问题的能力，RL 日益流行，甚至在与深度学习结合之前就已广泛应用。然而，深度学习与 RL 的结合不仅扩大了可解决问题的范围，也提高了成功率 [10]。这种结合在高维状态和动作空间中表现更佳。深度强化学习 (Deep RL) 已在多种应用中超越人类水平。深度 RL 在游戏领域的多项重要实验包括击败世界顶级扑克选手 [11, 12]、从像素级别玩视频游戏 [13, 14]，以及通过 AlphaGo 等程序掌握围棋 [15]。RL 和深度 RL 被广泛应用于学术和商业领域，包括机器人学 [16-19]、自动驾驶车辆 [20]、银行业 [21]、智能电网 [22] 和能源行业 [23]。在机器人学中，RL 使机器人通过试错与环境交互，学习最优路径或行为。RL 中的控制任务设计者无需明确定义问题解决的数学细节，而是通过标量目标函数形式提供反馈，衡量机器人单步表现。存在多种 RL 技术可用于解决连续体机器人的控制问题 [24-27]。例如，文献 [26] 采用基于深度 Q 学习的控制策略，成功实现二维位置到达。此外，文献 [27] 提出了一种不依赖特定模型的控制方法，利用 Q 学习，并将该方法应用于多段软体机械臂，实现 2D 位置到达。

RL can be categorized as either model-based or model-free. Two different methods are used in the model-free approach, which are value-based and policy-based. In policy-based methods, we directly train a policy function, while in value-based methods, we train a value function that maps a state to the expected value of being in that state. The foundations of these RL approaches and methods are based on the Markov decision process. The Markov decision process is an approach to predicting the best next state (s') by dynamically computing the rewards ($Q(s, a) = E[R[s, a]]$) of possible action options(a) for discrete-time scholastic events with respect to the current state(s).

强化学习可分为基于模型和无模型两类。无模型方法中采用两种不同策略：基于价值和基于策略。在基于策略的方法中，我们直接训练策略函数；而在基于价值的方法中，我们训练一个价值函数，将状态映射到该状态的期望价值。这些 RL 方法的基础是马尔可夫决策过程。马尔可夫决策过程是一种通过动态计算可能动作选项 (a) 在当前状态 (s) 下的奖励 ($Q(s, a) = E[R[s, a]]$)，以预测最佳下一状态 (s') 的离散时间随机事件方法。

The deep deterministic policy gradient (DDPG) algorithm intersects two different methods, combining the ideas of the deterministic policy gradient (DPG) and the deep Q-network [28]. DDPG consists of two deep neural networks, similar to the Actor-Critic method. The actor suggests an action given a state, while the critic predicts whether the action is good or bad with a continuous numeric value given a state and an action. Therefore, DDPG can solve the continuous state and action space. DDPG has become a frequently used algorithm in the literature [29-32]. For example, it has been used to stabilize the end-effector of a humanoid robot with 27 DOFs during its movement [29]. In another study, a specialized DDPG architecture called the twin-delayed DDPG structure was used to perform an inverse kinematic solution of a 5- DOF robot arm [30]. Detecting whether an object is in the workspace and touching an object in the workspace of a 7 DOF robot was achieved using a DDPG structure in another study [31]. Altitude control has been achieved not only in articulated robots, but also in systems such as unmanned aerial vehicles with the help of DDPG [32].

深度确定性策略梯度 (DDPG) 算法结合了确定性策略梯度 (DPG) 和深度 Q 网络 (Deep Q-Network, DQN)[28] 的思想。DDPG 由两个深度神经网络组成, 类似于 Actor-Critic 方法。Actor 根据状态提出动作建议, 而 Critic 则根据状态和动作预测该动作的优劣, 输出连续的数值。因此, DDPG 能够解决连续状态和动作空间的问题。DDPG 已成为文献中常用的算法 [29-32]。例如, 它被用于稳定具有 27 自由度 (DOFs) 的人形机器人末端执行器的运动 [29]。另一项研究中, 采用了一种称为双延迟 DDPG(twin-delayed DDPG) 的专用架构, 完成了 5 自由度机器人臂的逆运动学求解 [30]。还有研究利用 DDPG 结构实现了 7 自由度机器人工作空间内物体的检测和接触 [31]。DDPG 不仅在关节机器人中实现了高度控制, 还被应用于无人机等系统中 [32]。

This paper aims to demonstrate the implementation of a forward kinematic model using a constant curvature representation and a velocity kinematics framework to simulate the planar continuum robot environment. The objective is to solve a 2D position-reaching task using the RL approach to control a continuum robot in a simulation environment. The DDPG algorithm is employed to learn controls in continuous action spaces, which is crucial for the intended control mechanism. The algorithm's features were examined through several simulations, and the results are presented. The simulations involved the continuum robot reaching a goal state while minimizing distance error. To our knowledge, the combination of the DDPG algorithm with our chosen kinematic model is a unique contribution to the field. This research not only underscores the immediate applicability of our approach but also paves the way for its potential expansion to more complex robotic systems in the future.

本文旨在展示利用恒定曲率表示和速度运动学框架实现前向运动学模型, 以模拟平面连续机器人环境。目标是通过强化学习 (RL) 方法解决二维位置到达任务, 控制连续机器人在仿真环境中的运动。采用 DDPG 算法学习连续动作空间中的控制策略, 这对于预期的控制机制至关重要。通过多次仿真验证了该算法的特性, 并展示了结果。仿真过程中, 连续机器人在最小化距离误差的同时达到目标状态。据我们所知, 将 DDPG 算法与所选运动学模型结合应用是该领域的独特贡献。本研究不仅强调了该方法的即时适用性, 也为未来向更复杂机器人系统的扩展奠定了基础。

2 Kinematics Modeling of Continuum Robot

2 连续机器人运动学建模

Real-world applications of continuum robotics require the usage of robot shape and motion models. To simulate the control based on RL described in this paper, it's necessary to define the forward kinematic model, which provides the correct shape of the continuum robot, and the velocity kinematics, which provide the correct motion. This section provides an overview of forward and velocity kinematics design for continuum robots.

连续机器人在实际应用中需要使用机器人形状和运动模型。为了模拟本文所述的基于强化学习的控制, 必须定义前向运动学模型以提供连续机器人的正确形状, 以及速度运动学以提供正确的运动。本节概述了连续机器人的前向运动学和速度运动学设计。

The forward kinematics of continuum robots contains three spaces that can be decomposed into two mappings, which are a mapping between the robot's actuator space and the configuration space, and a mapping between the configuration space and the task space [1]. You can see the mapping in Fig. 2.

连续机器人的前向运动学包含三个空间，可分解为两个映射，即机器人执行器空间与配置空间之间的映射，以及配置空间与任务空间之间的映射 [1]。映射关系见图 2。

Several kinematic frameworks have been presented for continuum robot modeling. [33] discussed some of these models, such as variable curvature representation, constant curvature representation, pseudo-rigid body, and modal approach, in detail and shared the models implemented as open source. The planar constant curvature representation is chosen as the kinematic model in this paper due to its ease of implementation and a novel choice for the kinematic model in the context of RL-based continuum robot control. This simple model captures the key features of continuum robots, like their continuous structure and high degree of freedom. By focusing on the basics of continuum robot control, we could test our RL approach. The planar model is also relevant to real-world scenarios where continuum robots operate in two-dimensional spaces. The insights we gained from this study apply not only to the planar model, but also to more complex continuum robots.

已有多重连续机器人运动学框架被提出。[33] 详细讨论了其中一些模型，如变曲率表示、恒定曲率表示、伪刚体模型和模态方法，并分享了开源实现。本文选择平面恒定曲率表示作为运动学模型，因其易于实现且在基于强化学习的连续机器人控制中是一种新颖选择。该简单模型捕捉了连续机器人连续结构和高自由度的关键特征。通过聚焦连续机器人控制的基础，我们能够测试强化学习方法。该平面模型也适用于连续机器人在二维空间中运行的实际场景。本研究所得见解不仅适用于平面模型，也适用于更复杂的连续机器人。

2.1 Forward Kinematics with D-H Method

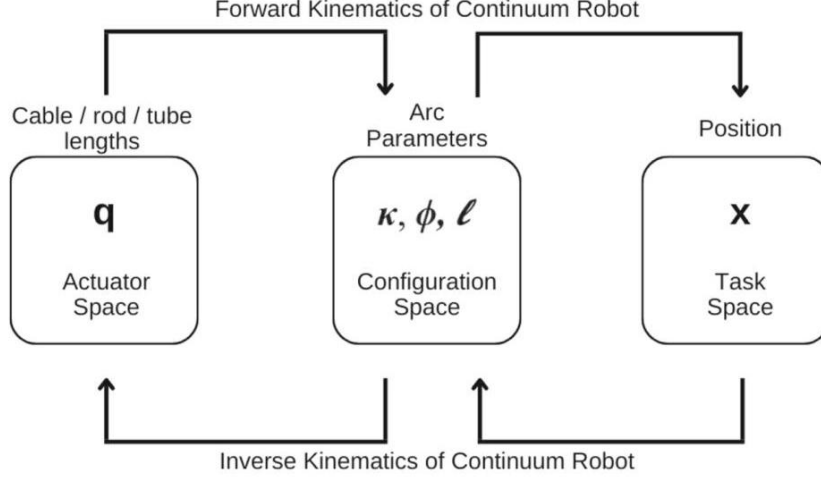
2.1 基于 D-H 方法的前向运动学

In this section of the paper, the solution for the geometric shape of a three-section continuum robot using the Denavit-Hartenberg (DH) method, which is one of the forward kinematics frameworks, is presented. The purpose of deriving the forward kinematics model of the continuum robot is to calculate the new position of the robot as a result of the actions taken by the robot in the control problem with the RL approach. In order to infer the shape of continuum designs, it is necessary to obtain a description of their movement. Because of the continuous nature of the design, joint angles and joint lengths do not provide a simple and physically feasible description of the manipulator's shape. Instead, a kinematic model that describes the manipulator's geometry using curvatures can offer a more realistic and understandable description. When using a constant curvature approach, a planar curve can be thought of as moving in three coupled movements.

本节介绍利用 Denavit-Hartenberg(D-H) 方法求解三段连续机器人几何形状的方案，该方法是前向运动学框架之一。推导连续机器人前向运动学模型的目的是计算机器人在强化学习控制问题中执行动作后的新位置。为了推断连续设计的形状，必须获得其运动描述。由于设计的连续性，关节角度和关节长度无法简单且物理可行地描述机械臂的形状。相反，使用曲率描述机械臂几何的运动学模型能提供更真实且易于理解的描述。采用恒定曲率方法时，平面曲线可视为在三个耦合运动中移动。

Fig. 2 Three space of

图 2 连续机器人三个空间及映射方向



continuum robot and the mapping directions

连续机器人及映射方向

1. rotation by an angle θ

1. 旋转角度为 θ

2. translation by an amount of $\|x\|$, and

2. 平移距离为 $\|x\|$, 以及

3. rotation by the angle θ once more,

3. 再次旋转角度为 θ

where $\mathbf{x}$ is referred to as the position vector of the curve's endpoint. With the above information, we have a discrete motion description of how the curve moves. [34] obtained the homogeneous transformation matrix in Eq. 1 using a modified D-H approach for forward kinematic analysis, just as is applied for conventional rigid joint manipulators.

其中 $\mathbf{x}$ 被称为曲线端点的位置向量。根据上述信息，我们得到了曲线运动的离散描述。[34] 采用改进的 D-H 方法获得了式 1 中的齐次变换矩阵，用于正向运动学分析，方法与传统刚性关节机械臂相同。

$$A_0^3 = A_0^1 A_1^2 A_2^3 = \begin{bmatrix} \cos(\kappa l) & -\sin(\kappa l) & 0 & \frac{1}{\kappa} \{\cos(\kappa l) - 1\} \\ \sin(\kappa l) & \cos(\kappa l) & 0 & \frac{1}{\kappa} \sin(\kappa l) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

The homogeneous transformation matrix for the curve from frame 0 to frame 3 can be written in terms of the curvature κ and the total arc length l . This representation gives us the shape of only the first section. In order to know the endpoint position of the three-section robot, the product of the matrix of each partition must be calculated.

The frames for a three-section continuum robot can be setup as a combination of three sections. The homogeneous transformation matrix used for forward kinematics is the product of three matrices of the form of Eq. 1. Therefore, the transformation matrix from frame 0 to frame 9 is shown in Eq. 2.

从坐标系 0 到坐标系 3 的曲线齐次变换矩阵可以用曲率 κ 和总弧长 l 表示。该表示仅给出了第一段的形状。为了确定三段机器人末端位置，必须计算各分段矩阵的乘积。三段连续体机器人的坐标系可以视为三段的组合。用于正向运动学的齐次变换矩阵是三个形式如式 1 的矩阵的乘积。因此，从坐标系 0 到坐标系 9 的变换矩阵如式 2 所示。

$$A_0^9 = A_0^3 A_3^6 A_6^9$$

$$= \begin{bmatrix} \cos(\omega_1 + \omega_2 + \omega_3) & -\sin(\omega_1 + \omega_2 + \omega_3) & 0 & A_{14} \\ \sin(\omega_1 + \omega_2 + \omega_3) & \cos(\omega_1 + \omega_2 + \omega_3) & 0 & A_{24} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where κ_i and l_i are the curvature and total arc length, respectively for section i and $\omega_i = \kappa_i l_i$ for $i = \{1, 2, 3\}$. The following are the elements of the matrix that were not previously defined:

其中 κ_i 和 l_i 分别是第 i 段的曲率和总弧长， $\omega_i = \kappa_i l_i$ 和 $i = \{1, 2, 3\}$ 分别对应 $i = \{1, 2, 3\}$ 。以下是之前未定义的矩阵元素：

$$A_{14} = \frac{1}{\kappa_1} \{\cos \omega_1 - 1\} + \frac{1}{\kappa_2} \{\cos(\omega_1 + \omega_2) - \cos \omega_1\}$$

$$+ \frac{1}{\kappa_3} \{\cos(\omega_1 + \omega_2 + \omega_3) - \cos(\omega_1 + \omega_2)\} \quad (3)$$

$$A_{24} = \frac{1}{\kappa_1} \sin \omega_1 + \frac{1}{\kappa_2} \{\sin(\omega_1 + \omega_2) - \sin \omega_1\}$$

$$+ \frac{1}{\kappa_3} \{\sin(\omega_1 + \omega_2 + \omega_3) - \sin(\omega_1 + \omega_2)\}$$

Equation 3 captures the spatial relationships and transformations as the continuum robot moves in the planar space. The components A_{14} and A_{24} of the transformation matrix are derived from the robot's configuration and kinematic constraints. A_{14} represents the x-coordinate transformation and A_{24} represents the y-coordinate transformation. Note that the total arc length for $i = \{1, 2\}$ must be used so that Section 3 is properly positioned, but any arc length can be used for the final section depending on where the point of interest lies in the section. The total arc length for the final section gives the kinematics in terms of the end point. To provide a clearer understanding of the planar continuum robot used in our study, we present the kinematics parameters in Fig. 3 for a single section of the continuum robot. r defines the radius of the resulting arc, which is equal to $\frac{1}{\kappa}$.

式 3 描述了连续体机器人在平面空间运动时的空间关系和变换。变换矩阵的分量 A_{14} 和 A_{24} 来源于机器人的构型和运动学约束。 A_{14} 表示 x 坐标变换， A_{24} 表示 y 坐标变换。注意，为了正确定位第 3 段，必须使用其总弧长 $i = \{1, 2\}$ ，但对于最终段，可以根据感兴趣点在该段的位置使用任意弧长。最终段的总弧长给出了末端点的运动学。为更清晰地理解本研究中使用的平面连续体机器人，我们在图 3 中展示了单段连续体机器人的运动学参数。 r 定义了生成弧的半径，等于 $\frac{1}{\kappa}$ 。

2.2 Velocity Kinematics

2.2 速度运动学

Positions are not enough to create movement. Therefore, velocities are needed for a better interaction. Given curvature velocities, we can calculate how fast the endpoint moves and how fast each curvature must move to guarantee a desired endpoint velocity using the velocity kinematics framework.

仅有位置不足以产生运动，因此需要速度以实现更好的交互。给定曲率速度，我们可以计算末端点的移动速度以及各曲率必须移动的速度，以保证期望的末端速度，这基于速度运动学框架。

The velocity kinematics of a continuum robot can be expressed similar to traditional kinematic analysis as Eq. 4

连续体机器人的速度运动学可以类似传统运动学分析表示，如式 4 所示

$$\dot{x} = J\dot{q} \quad (4)$$

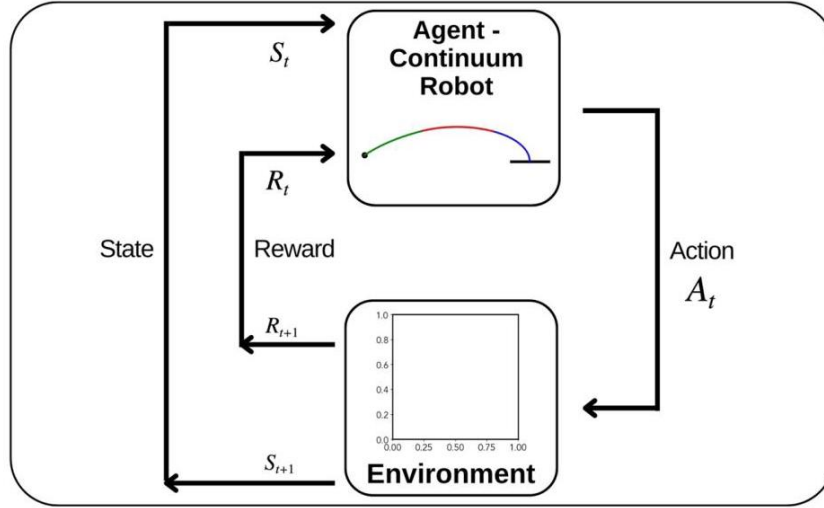
where $x \in \mathbb{R}^{m \times 1}$ is the vector of the task space that corresponds to position and / or orientation, and the dot notation represents differentiation with respect to time, denoting the time derivative of the position and/or orientation variables. For the sake of simplicity, we have not taken orientation into account in our current work. While orientation is an important aspect of robot control, our primary focus in this study is on position control. By simplifying the control task to exclude orientation, we aim to establish a foundational understanding of the robot's planar movement dynamics using our RL approach. However, we acknowledge that in real-world applications, orientation can play a crucial role, and we plan to incorporate it in our future work to provide a more comprehensive control solution. To be clear for our three-section planar robot:

其中 $x \in \mathbb{R}^{m \times 1}$ 是对应位置和/或姿态的任务空间向量，点号表示对时间的微分，即位置和/或姿态变量的时间导数。为简化起见，本研究未考虑姿态。虽然姿态是机器人控制的重要方面，但本研究主要关注位置控制。通过简化控制任务排除姿态，我们旨在利用强化学习方法建立机器人平面运动动力学的基础理解。然而，我们承认在实际应用中姿态可能起关键作用，计划在未来工作中纳入姿态以提供更全面的控制方案。针对我们的三段平面机器人，说明如下：

$$\dot{\mathbf{q}} = [\dot{\kappa}_1, \dot{\kappa}_2, \dot{\kappa}_3]^T, \dot{\mathbf{x}} = [\dot{x}, \dot{y}]^T \quad (5)$$

Fig. 3 Illustration of specified

图 3 指定的示意图



kinematics parameters

运动学参数

The matrix J called as Jacobian matrix, and is a function of the 'joint' variables $\mathbf{q}$. The Jacobian matrix represents the linear sensitivity of the tip point of the continuum robot velocity $\dot{x}$ to the curvature velocity $\dot{\kappa}$, and it is a function of curvature variables κ . When taking partial derivatives of forward kinematic equations, as in Eq. 6, one can obtain the components of the Jacobian J_{ij} :

矩阵 J 称为雅可比矩阵，是‘关节’变量 $\mathbf{q}$ 的函数。雅可比矩阵表示连续体机器人末端速度 $\dot{x}$ 对曲率速度 $\dot{\kappa}$ 的线性敏感性，且是曲率变量 κ 的函数。通过对正向运动学方程 (如式 6) 求偏导，可以得到雅可比矩阵的分量 J_{ij} ：

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} \quad (6)$$

Equation 7 will result from writing it using the chains rule.

式 7 是利用链式法则写出的结果。

$$\begin{aligned} \frac{dx}{dt} &= \frac{\partial x}{\partial q_1} \frac{dq_1}{dt} + \frac{\partial x}{\partial q_2} \frac{dq_2}{dt} \\ \frac{dy}{dt} &= \frac{\partial y}{\partial q_1} \frac{dq_1}{dt} + \frac{\partial y}{\partial q_2} \frac{dq_2}{dt} \end{aligned} \quad (7)$$

If this case in Eq. 7 applied to the continuum robot, Eq. 8 is obtained.

若将式 7 应用于连续体机器人，则得到式 8。

$$\frac{dx}{dt} = \frac{\partial x}{\partial \kappa_1} \frac{d\kappa_1}{dt} + \frac{\partial x}{\partial \kappa_2} \frac{d\kappa_2}{dt} + \frac{\partial x}{\partial \kappa_3} \frac{d\kappa_3}{dt}$$

$$\frac{dy}{dt} = \frac{\partial y}{\partial \kappa_1} \frac{d\kappa_1}{dt} + \frac{\partial y}{\partial \kappa_2} \frac{d\kappa_2}{dt} + \frac{\partial y}{\partial \kappa_3} \frac{d\kappa_3}{dt} \quad (8)$$

We can extract the Jacobian Matrix elements as below:

我们可以提取雅可比矩阵的元素如下:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \kappa_1} & \frac{\partial x}{\partial \kappa_2} & \frac{\partial x}{\partial \kappa_3} \\ \frac{\partial y}{\partial \kappa_1} & \frac{\partial y}{\partial \kappa_2} & \frac{\partial y}{\partial \kappa_3} \end{bmatrix} \quad (9)$$

Finally, we obtain Eq. 10.

最终得到式 10。

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \kappa_1} & \frac{\partial x}{\partial \kappa_2} & \frac{\partial x}{\partial \kappa_3} \\ \frac{\partial y}{\partial \kappa_1} & \frac{\partial y}{\partial \kappa_2} & \frac{\partial y}{\partial \kappa_3} \end{bmatrix} \begin{bmatrix} \frac{d\kappa_1}{dt} \\ \frac{d\kappa_2}{dt} \\ \frac{d\kappa_3}{dt} \end{bmatrix} \quad (10)$$

Robotics and control theory often rely on Jacobian matrices. In short, a Jacobian helps to explain how two different system representations interact dynamically. In our specific example, this connection is based on both position and curvature. The aim of velocity kinematics is to describe the motion of an endpoint. However, the motion of the endpoint is heavily restricted by curvatures. As a result, describing the motion of the endpoint also means describing the states of motion at those curvatures.

机器人学和控制理论常常依赖雅可比矩阵。简而言之，雅可比矩阵有助于解释两种不同系统表示之间的动态交互。在我们的具体示例中，这种联系基于位置和曲率。速度运动学的目标是描述端点的运动。然而，端点的运动受到曲率的强烈限制。因此，描述端点的运动也意味着描述这些曲率处的运动状态。

The velocity kinematics plays a pivotal role in generating movement. In our approach, the action is defined as the derivative of curvature with respect to time ($\dot{\kappa}$). By incorporating the Jacobian matrix derived from the forward kinematics, we are able to calculate the movement at each time step, thus facilitating a smooth and controlled movement of the robot. This integration of velocity kinematics is crucial in creating a dynamic and responsive control system.

速度运动学在生成运动中起着关键作用。在我们的方法中，动作定义为曲率关于时间的导数 ($\dot{\kappa}$)。通过结合由正向运动学导出的雅可比矩阵，我们能够计算每个时间步的运动，从而实现机器人的平滑且受控的运动。速度运动学的这种整合对于创建动态且响应迅速的控制系统至关重要。

3 Reinforcement Learning

3 强化学习

Both humans and animals learn best through trial and error. They interact with their surroundings, make small changes, and observe the outcomes. For example, consider how a newborn learns to walk. It may take a few days of attempting different actions before they can stand up and walk without falling. During this process, the baby learns

from rewards and penalties - they are rewarded for making progress and penalized for falling [35]. This incentive system is innate to humans and animals and encourages us to pursue behaviors that result in positive rewards and avoid behaviors that result in negative ones. This type of learning is called reinforcement learning and is modeled mathematically based on human and animal learning patterns. In a reinforcement learning framework, an agent interacts with the environment E by receiving observations s_t and executing actions a_t . After each t time step, the agent receives a scalar reward r_t and a new state s_{t+1} . Figure 4 illustrates this process.

人类和动物都通过反复试验学习效果最佳。他们与环境互动，做出小的调整，并观察结果。例如，考虑新生儿如何学走路。可能需要几天时间尝试不同动作，才能站立并行走而不摔倒。在此过程中，婴儿通过奖励和惩罚学习——他们因取得进步而获得奖励，因跌倒而受到惩罚 [35]。这种激励机制是人类和动物的天性，促使我们追求带来正向奖励的行为，避免带来负面结果的行为。这种学习方式称为强化学习，其数学模型基于人类和动物的学习模式。在强化学习框架中，智能体通过接收观察 E 并执行动作 s_t 与环境交互 a_t 。每个 t 时间步后，智能体获得一个标量奖励 r_t 和一个新状态 s_{t+1} 。图 4 展示了这一过程。

Algorithms for reinforcement learning can be broadly divided into two groups. Model-based and model-free approaches are the names for these. Model-based algorithms are constructed using the environment's prior knowledge or a mathematical model that represents it. Algorithms for model-free reinforcement learning are unaware of their environment. By simulating the environment with a Markov decision process, which is typically a random process, algorithms are created and learned. The agent investigates the environment, acquires knowledge about them, and then chooses the best course of action. This procedure is an example of reinforcement learning that involves trial and error as well as reward accumulation. Today, reinforcement learning is used in many different fields, and one of the key factors in its success is that it doesn't need an environment model. It interacts with the environment in this way to find solutions to a variety of engineering issues. However, it has often been used in robotics and control problems.

强化学习算法大致可分为两类：基于模型和无模型方法。基于模型的算法利用环境的先验知识或其数学模型构建。无模型强化学习算法则对环境一无所知。通过用马尔可夫决策过程（通常是随机过程）模拟环境，算法得以创建和学习。智能体探索环境，获取知识，然后选择最佳行动方案。该过程是强化学习中试错和奖励积累的典型例子。如今，强化学习被广泛应用于多个领域，其成功的关键之一是无需环境模型。它通过与环境的交互，寻找解决各种工程问题的方案。然而，它常被用于机器人学和控制问题。

Basic RL algorithms such as Q learning and Sarsa learning are called tabular methods in the literature [37]. For discrete state and action spaces, these techniques are appropriate. The control problem's state space and the control signal are actually stated in continuous time. The derivative values of the curvature, which is our control signal, in the continuum robot control problem discussed in this paper are defined continuous. By discretizing the continuous state and action spaces, Q learning and Sarsa learning algorithms can be used as practical implementations of reinforcement learning, however in reality, the errors caused by discretization constitute a concern in control applications. Another problem with the tabular method of reinforcement learning is the computational cost. Increasing the size of our discrete state and action space leads to an increase in computational complexity and computational cost.

基本的强化学习算法如 Q 学习和 Sarsa 学习在文献中被称为表格方法 [37]。这些方法适用于离散的状态和动作空间。控制问题的状态空间和控制信号实际上是连续时间定义的。本文讨论的连续体机器人控制问题中，作为控制信号的曲率导数值是连续定义的。通过对连续状态和动作空间进行离散化，可以将 Q 学习和 Sarsa 学习算法作为强化学习的实际实现，但实际上，离散化带来的误差在控制应用中是一个问题。表格方法的另一个问题是计算成本。随着离散状态和动作空间规模的增加，计算复杂度和计算成本也随之上升。

As the amount of accessible data grows exponentially every year and the hardware quality of computers improves (especially GPUs), deep learning methods have started to develop rapidly, and thus deep reinforcement learning was born from the combination of artificial neural networks with reinforcement learning [10]. Thus, algorithms that can operate in continuous state and action space have been developed. The application of RL to the control of continuum robots offers several potential advantages and disadvantages compared to other existing control methods. Pros of RL for Continuum Robot Control as follows:

随着可用数据量每年呈指数增长以及计算机硬件性能 (尤其是 GPU) 的提升，深度学习方法开始迅速发展，因而深度强化学习诞生于人工神经网络与强化学习的结合 [10]。因此，能够在连续状态和动作空间中运行的算法得以开发。将强化学习应用于连续体机器人控制相比其他现有控制方法具有若干潜在优缺点。连续体机器人控制中强化学习的优点如下：

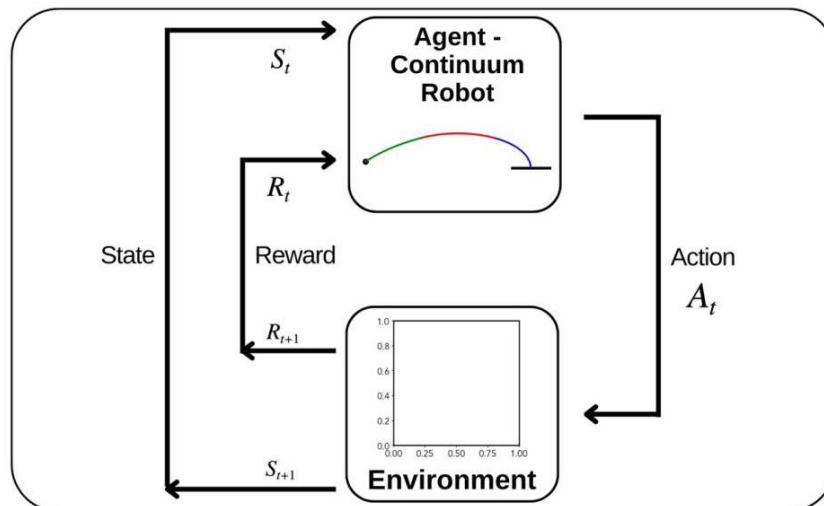
- Adaptability to changing environments and system dynamics.
- 适应环境变化和系统动力学的能力。
- Flexibility for complex and nonlinear systems.
- 适用于复杂和非线性系统的灵活性。
- Autonomous learning without relying on human-engineered controllers.
- 无需依赖人工设计的控制器，实现自主学习。
- Scalability to high-dimensional state and action spaces.
- 可扩展至高维状态和动作空间。
- Generalization to similar tasks or robot platforms.
- 对类似任务或机器人平台的泛化能力。

Cons of RL for Continuum Robot Control as follows:

连续体机器人控制中强化学习的缺点如下：

Fig. 4 Agent and Environment interaction in Reinforcement Learning

图 4 强化学习中智能体与环境的交互



- Sample inefficiency, requiring numerous interactions with the environment.

• 样本效率低，需要大量与环境的交互。

- Exploration-exploitation trade-off, challenging in continuous and flexible systems.

• 探索与利用的权衡，在连续且柔性的系统中具有挑战性。

- Safety concerns due to trial and error learning.

• 由于试错学习带来的安全隐患。

- Hyperparameter tuning for optimal performance.

• 超参数调优以实现最佳性能。

- Lack of interpretability in learned policies.

• 学习策略缺乏可解释性。

In conclusion, while RL offers potential advantages for the control of continuum robots in terms of adaptability, flexibility, autonomy, scalability, and generalization, it also has limitations in terms of sample efficiency, exploration-exploitation trade-off, safety concerns, hyperparameter tuning, and interpretability. Careful consideration of these pros and cons is necessary when applying RL to the control of continuum robots, and further research and development are needed to address the challenges and improve the performance and safety of RL-based control methods for continuum robots.

总之，尽管强化学习 (RL) 在连续体机器人控制方面具有适应性、灵活性、自主性、可扩展性和泛化能力等潜在优势，但在样本效率、探索与利用的权衡、安全性问题、超参数调优以及可解释性方面也存在局限。在将强化学习应用于连续体机器人控制时，需要仔细权衡这些利弊，并且需要进一步的研究和开发以解决这些挑战，提升基于强化学习的连续体机器人控制方法的性能和安全性。

3.1 Deep Reinforcement Learning

3.1 深度强化学习

Deep reinforcement learning combines deep neural networks with reinforcement learning algorithms to solve continuous state control problems. While table-based algorithms like Q-learning do not work well for these types of problems, deep reinforcement learning methods are often used. One popular algorithm is deep Q-learning, but it is not effective for continuous action space due to its reliance on a single neural network. To address this issue, new algorithms with two neural networks that work well for both continuous state space and continuous action space have been introduced [28, 38, 39].

深度强化学习结合了深度神经网络与强化学习算法，用于解决连续状态控制问题。表格型算法如 Q 学习不适用于此类问题，因此通常采用深度强化学习方法。一种流行的算法是深度 Q 学习，但由于其依赖单一神经网络，对于连续动作空间效果不佳。为解决此问题，提出了采用两个神经网络的新算法，能够同时适用于连续状态空间和连续动作空间 [28, 38, 39]。

As mentioned earlier, the environment of the continuum robot is characterized by continuous control action and state. This is an important factor to consider when choosing a reinforcement learning algorithm. To address this issue, we have employed the DDPG algorithm, which is capable of handling continuous action and state space. The DDPG algorithm uses neural networks approximations to represent the actor-critic algorithm. Based on the policy π , which is determined by the present state, the actor chooses the current action. The critic takes the current state and action as input, and is described by an action-value function Q . In this paper, a deep neural network with several layers is used to approximate the functions Q and π .

如前所述，连续体机器人的环境以连续控制动作和状态为特征。这是选择强化学习算法时需考虑的重要因素。为此，我们采用了 DDPG 算法，该算法能够处理连续动作和状态空间。DDPG 算法使用神经网络近似表示演员-评论家 (actor-critic) 算法。基于当前状态确定的策略 π ，演员选择当前动作。评论家以当前状态和动作为输入，由动作价值函数 Q 描述。本文中，采用多层深度神经网络来近似函数 Q 和 π 。

A DDPG algorithm is a model-free off-policy algorithm formed by combining the deterministic policy gradient and deep Q Learning. It is built on DPG, which can operate over continuous action spaces, and uses Experience Replay and slow-learning target networks from deep Q learning. To give a detailed mathematical background of the DDPG algorithm, robot performs an action $a_t \in \mathbb{R}^N$ in the environment where the agent is in discrete time steps and records s_{t+1}, r_t . The aim of the agent is to maximize the sum of all rewards $R_t = \sum_{t=1}^T \gamma^{(t-1)} r_t(s_t, a_t)$ that it will create in the future. Depending on our π policy, the action we take in a given state and the reward values we will obtain accordingly change. The expression of the policy can be deterministic or stochastic. Equation 11 describes the expected return of taking an action a_t in a certain state s_t , expressed by following a policy π . This return is represented by the action value function [28].

DDPG 算法是一种无模型的离策略算法，由确定性策略梯度 (deterministic policy gradient) 和深度 Q 学习结合而成。它基于 DPG，能够在连续动作空间中运行，并借鉴了深度 Q 学习中的经验回放 (Experience Replay) 和缓慢更新的目标网络。为了详细介绍 DDPG 算法的数学背景，机器人在环境中执行动作 $a_t \in \mathcal{A}$ ，代理以离散时间步进行操作并记录 s_{t+1}, r_t 。代理的目标是最大化未来所有奖励 $R_t = \sum_{t=1}^T \gamma^{(t-1)} r_t(s_t, a_t)$ 的总和。根据我们的策略 π ，在给定状态下采取的动作及相应获得的奖励值会发生变化。策略表达可以是确定性的也可以是随机的。公式 11 描述了在某状态 s_t 下采取动作 a_t 并遵循策略 π 时的期望回报，该回报由动作价值函数表示 [28]。

$$Q^\pi(s_t, a_t) = E[R_t | s_t, a_t] \quad (11)$$

Extending Eq. 11 with the Bellman equation as a recursive approximation, we obtain Eq. 12 for deterministic policies.

将公式 11 与贝尔曼方程作为递归近似扩展，得到确定性策略的公式 12。

$$Q^\pi(s_t, a_t) = E[r(s_t, a_t) + \gamma Q^\pi(s_{t+1}, \pi(s_{t+1}))] \quad (12)$$

The $\pi(s_{t+1})$ policy in the off-policy algorithms like Q learning algorithm is the ϵ -greedy approach. Parametric writing of Eq. 12 to minimize the loss is shown in Eq. 13.

在离策略算法如深度 Q 学习算法中，策略 $\pi(s_{t+1})$ 采用贪婪策略 ϵ 。公式 12 的参数化表达以最小化损失，见公式 13。

$$L(\theta^Q) = E[(Q(s_t, a_t; \theta^Q) - y_t)^2] \quad (13)$$

where

其中

$$y_t = r(s_t, a_t) + \gamma Q(s_{t+1}, \pi(s_{t+1}); \theta^Q) \quad (14)$$

It is possible to approximate Eq. 13 with multi-layer neural networks. Equation 13 cannot be calculated in high-dimensional action space with the Q learning algorithm. The actor-critic approach is used to solve this problem. The deterministic policy gradient maps states to actions using a deterministic parametric policy function $\pi(s | \theta^\mu)$. Using the Critic $Q(s, a)$ Bellman equation, Q is calculated as in learning. The actor updates his parameters using a gradient-based update using Eq. 15.

可以用多层神经网络近似公式 13。公式 13 无法用深度 Q 学习算法在高维动作空间中计算。为解决此问题，采用演员-评论家方法。确定性策略梯度使用确定性参数策略函数 $\pi(s | \theta^\mu)$ 将状态映射到动作。利用评论家 $Q(s, a)$ 的贝尔曼方程，动作价值函数 Q 按深度 Q 学习方式计算。演员通过基于梯度的更新，使用公式 15 更新其参数。

$$\nabla_{\theta^\pi} J = E[\nabla_a Q(s, a | \theta^Q) | s = s_t,$$

$$a = \pi(s_t) \nabla_{\theta^\pi} \pi(s_t | \theta^\pi) | s = s_t] \quad (15)$$

Equation 15 is called the policy gradient. The $\pi(s_t | \theta^\pi)$ function is created with the deterministic policy gradient algorithm and the deep Q learning algorithm approach. Deep neural networks are used to approximate the $\pi(s_t | \theta^\pi)$ function. The values $D = (s_t, a_t, r_t, s_{t+1})$ are saved in the created memory. The s_t states are the current position and goal position values for our robot. These different quantitative characteristics in the data cause different scaling. By providing normalization of the data formed in the memory, it is ensured that the artificial neural networks to be trained optimize their parameters. The biggest challenge in continuous action space is to discover the environment in which the agent is located by applying appropriate actions. Since the DDPG algorithm is an off-policy, it allows us to create the discovery problem independently of the algorithm. To provide exploration, an Ornstein-Uhlenbeck process is added to generate noise to the policy value produced by the actor, as demonstrated in Eq. 16.

方程 15 称为策略梯度。 $\pi(s_t | \theta^\pi)$ 函数是通过确定性策略梯度算法 (Deterministic Policy Gradient) 和深度 Q 学习算法方法构建的。深度神经网络用于近似 $\pi(s_t | \theta^\pi)$ 函数。数值 $D = (s_t, a_t, r_t, s_{t+1})$ 被保存在创建的内存中。 s_t 状态是我们机器人当前的位置和目标位置值。数据中这些不同的量化特征导致不同的缩放。通过对记忆中形成的数据进行归一化，确保待训练的人工神经网络优化其参数。连续动作空间中最大的挑战是通过执行适当的动作发现智能体所处的环境。由于 DDPG 算法是离策略 (off-policy) 的，它允许我们独立于算法创建探索问题。为了提供探索，向由执行者产生的策略值中添加了 Ornstein-Uhlenbeck 过程以生成噪声，如方程 16 所示。

$$\pi'(s_t) = \pi(s_t | \theta_t^\pi) + N \quad (16)$$

Table 1 Noise parameters of the DDPG Algorithm

表 1 DDPG 算法的噪声参数

Parameter	Value
μ	0.00
θ	0.15
σ	0.20

参数	数值
μ	0.00
θ	0.15
σ	0.20

You can find the parameters of the Ornstein-Uhlenbeck process in Table 1.

你可以在表 1 中找到 Ornstein-Uhlenbeck 过程的参数。

where μ represents the mean value of the process, θ is a positive constant that determines the speed of mean reversion, and σ is a positive constant that determines the volatility of the process.

其中 μ 表示过程的均值， θ 是决定均值回复速度的正常数， σ 是决定过程波动性的正常数。

3.1.1 Network Structure of the Reinforcement Learning Approach

3.1.1 强化学习方法的网络结构

In our endeavor to develop a robust control mechanism for planar continuum robots, we have adopted the DDPG algorithm, a model-free, off-policy actor-critic algorithm that operates in continuous action spaces. This section delineates the structure of the actor and critic networks and the synergies between them, offering a comprehensive insight into our RL approach. The RL approach is grounded on a dual neural network architecture comprising two primary components: the Actor network and the Critic network. This architecture is designed to seamlessly integrate the continuous control capabilities of the actor network with the value estimation proficiencies of the critic network, fostering a harmonized learning environment.

为了开发一种稳健的平面连续体机器人控制机制，我们采用了 DDPG 算法，这是一种无模型、离策略的演员-评论家算法，适用于连续动作空间。本节描述了演员网络和评论家网络的结构及其协同作用，全面介绍了我们的强化学习方法。该方法基于双神经网络架构，包含两个主要部分：演员网络和评论家网络。该架构旨在无缝整合演员网络的连续控制能力与评论家网络的价值估计能力，促进协调的学习环境。

The Actor network, responsible for determining the optimal policy, is structured as a feedforward neural network. It maps the current state of the robot to a specific action, aiming to maximize the expected cumulative reward. The network consists of:

演员网络负责确定最优策略，结构为前馈神经网络。它将机器人的当前状态映射到特定动作，旨在最大化期望累积奖励。该网络包括：

- Input Layer: Accepts the state variables, initiating the process of policy determination.
• 输入层：接受状态变量，启动策略确定过程。
- Hidden Layers: Incorporates four hidden layers with 128, 256, 256, and 128 neurons respectively, facilitating the learning of complex patterns in the data.
• 隐藏层：包含四个隐藏层，分别有 128、256、256 和 128 个神经元，促进对数据中复杂模式的学习。
- Output Layer: Utilizes a tanh activation function to generate actions within a specified range, aligning with the continuous action space of the problem.
• 输出层：使用 tanh 激活函数生成指定范围内的动作，符合问题的连续动作空间。

The Critic network, on the other hand, evaluates the value of the chosen action in the current state, aiding in the optimization of the policy developed by the actor network. Its structure is as follows:

评论家网络则评估当前状态下所选动作的价值，辅助优化演员网络制定的策略。其结构如下：

- Input Layer: Receives both the state variables and the actions proposed by the actor network as inputs.

- 输入层: 接收状态变量和演员网络提出的动作作为输入。

- Hidden Layers: Comprises four hidden layers with a similar configuration to the actor network, enhancing the ability to learn intricate relationships between states and actions.

- 隐藏层: 包含四个隐藏层, 配置与演员网络相似, 增强学习状态与动作间复杂关系的能力。

- Output Layer: Produces a single output representing the estimated Q-value of the chosen action-state pair.

- 输出层: 输出单一值, 表示所选动作-状态对的估计 Q 值。

ReLU (Rectified Linear Unit) activation functions are employed in the hidden layers of both networks, promoting efficient learning by mitigating the vanishing gradient problem. The learning process is facilitated through the backpropagation algorithm, complemented by the Adam optimizer, fostering efficient convergence during the training phase. Training involves minimizing a loss function that represents the difference between the predicted and target Q-values, a process orchestrated by the critic network. To prevent overfitting and foster model generalization, batch normalization and dropout techniques have been implemented. A set of optimized hyperparameters, including actor learning rate of 0.0001, critic learning rate of 0.0003 and a discount factor of 0.99, were determined through preliminary experiments to ensure stable and efficient learning. The performance of the networks was assessed based on convergence speed and the precision in achieving the desired goal states in the simulation environment. Further insights into performance metrics are elucidated in the "Simulations and results" section. The Table 2 displays the hyperparameters to summarize the discussion in this section.

两个网络的隐藏层均采用 ReLU(修正线性单元) 激活函数, 促进高效学习, 缓解梯度消失问题。学习过程通过反向传播算法实现, 辅以 Adam 优化器, 促进训练阶段的高效收敛。训练目标是最小化预测 Q 值与目标 Q 值之间的损失函数, 由评论家网络协调执行。为防止过拟合并促进模型泛化, 采用了批量归一化和丢弃技术。通过初步实验确定了一组优化超参数, 包括演员学习率 0.0001、评论家学习率 0.0003 和折扣因子 0.99, 以确保学习的稳定性和效率。网络性能基于收敛速度及在仿真环境中实现期望目标状态的精度进行评估。更多性能指标详见“仿真与结果”部分。表 2 展示了 DDPG 算法的超参数, 以总结本节内容。

Table 2 Hyperparameters of the DDPG Algorithm

表 2 DDPG 算法的超参数

Parameter	Value
Optimizer	Adam
Learning Rate Actor	0.0001
Learning Rate Critic	0.0003
Discount Factor (γ)	0.99
Batch Size	128
Replay Buffer Size	1000000
Actor Hidden Layer Number	4
Actor Hidden Layer Activation Function	Relu
Actor Output Layer Activation Function	Tangent Hyperbolic
Critic Hidden Layer Number	4
Critic Hidden Layer Activation Function	Relu
Critic Output Layer Activation Function	Linear
Maximum training steps	1000
Soft Update (τ)	0.001

参数	数值
优化器	Adam
演员学习率	0.0001
评论家学习率	0.0003
折扣因子 (γ)	0.99
批量大小	128
重放缓冲区大小	1000000
演员隐藏层数量	4
演员隐藏层激活函数	Relu
演员输出层激活函数	双曲正切
评论家隐藏层数量	4
评论家隐藏层激活函数	Relu
评论家输出层激活函数	线性
最大训练步数	1000
软更新 (τ)	0.001

3.2 Environment Design

3.2 环境设计

The interaction between the environment, actors, and critics is well shown in the DDPG diagram in Fig. 5. In DDPG based reinforcement learning, the environment responds to an actor's action by sending the critic an observation along with a reward. The environment's internal state encompasses all observations made, and the reward reflects the action's success. To utilize this approach, two elements are required. The first is the actor and critic networks, which are two distinct deep neural networks that determine the action and Q value for the robot. The second is the environment, which is a task or simulation that the Actor-Critic must solve in the robot's task space. This section provides details on our design of the environment for implementing the DDPG algorithm, which is an essential component of the process.

环境、执行者 (actor) 和评论者 (critic) 之间的交互在图 5 的 DDPG 示意图中得到了充分展示。在基于 DDPG 的强化学习中，环境通过向评论者发送观察值和奖励来响应执行者的动作。环境的内部状态包含所有观察值，奖励反映动作的成功程度。要利用这种方法，需要两个要素。第一个是执行者和评论者网络，它们是两个独立的深度神经网络，用于确定机器人动作和 Q 值。第二个是环境，即执行者-评论者必须在机器人任务空间中解决的任务或仿真。本节详细介绍了我们为实现 DDPG 算法设计的环境，这是该过程的关键组成部分。

The aim of this task is to achieve the 2D position reach by moving a three-section continuum robot from its initial state q_{ini} to the goal state q_{goal} . The initial position of the robot, denoted as (x_{ini}, y_{ini}) , and the target position, denoted as (x_{goal}, y_{goal}) , are randomly selected within the task space, which is illustrated in Fig. 7. The task space is defined for a continuum robot with a section length of 0.1 [m] and curvature values ranging from $-4 \text{ [}\frac{1}{m}\text{]}$ to $16 \text{ [}\frac{1}{m}\text{]}$. Additionally, the base of the continuum robot is mounted at the point $(0,0)$.

该任务的目标是通过移动一个三节连续体机器人，从其初始状态 q_{ini} 达到目标状态 q_{goal} ，实现 2D 位置的到达。机器人的初始位置记为 (x_{ini}, y_{ini}) ，目标位置记为 (x_{goal}, y_{goal}) ，均在任务空间内随机选取，如图 7 所示。任务空间定义了连续体机器人，其节长为 0.1 [m] ，曲率值范围从 $-4 \text{ [}\frac{1}{m}\text{]}$ 到 $16 \text{ [}\frac{1}{m}\text{]}$ 。此外，连续体机器人的基座安装在点 $(0,0)$ 处。

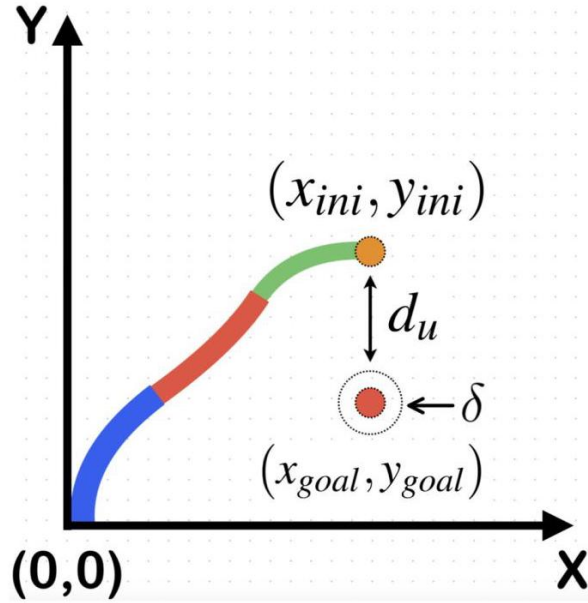


Fig. 6 Three-section continuum robot with initial (orange) and goal (red) state, goal tolerance δ . First section (blue), middle section (red) and third section (red)

图 6 三节连续体机器人，初始状态 (橙色) 和目标状态 (红色)，目标容差 δ 。第一节 (蓝色)、中间节 (红色) 和第三节 (红色)

Figure 6 is included in the paper to provide readers with a visual representation of the task that the robot is required to perform. In this figure, d_u represents the Euclidean distance between the robot's current state and the target point, while δ represents the tolerance or target circle that the robot must enter in order to reach the target. This graphical representation helps readers better understand the criteria for successful completion of the task (Fig. 7).

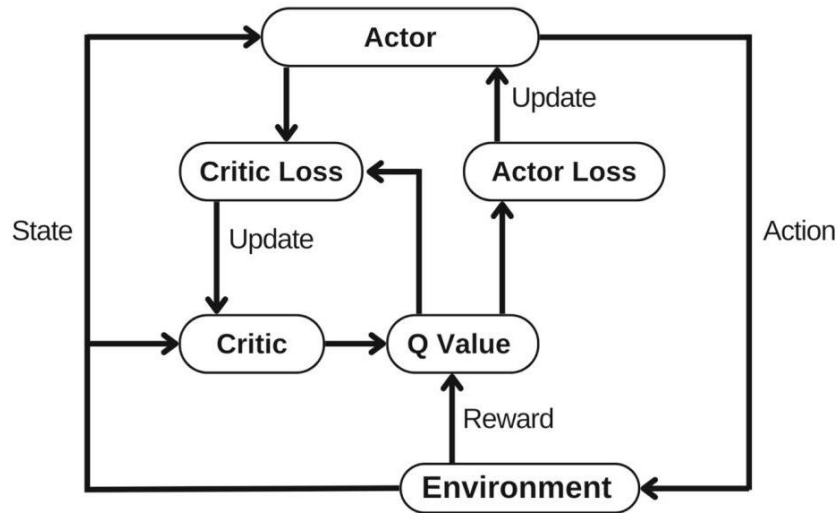
图 6 包含在论文中，旨在为读者提供机器人需执行任务的直观展示。图中， d_u 表示机器人当前状态与目标点之间的欧氏距离， δ 表示机器人必须进入的容差或目标圆，以达到目标。该图形表示有助于读者更好地理解任务成功完成的标准 (见图 7)。

The environment specification is a crucial component of the reinforcement learning methodology. In our study, we directly apply the forward and velocity kinematics of the robot, as covered in Section 2, to describe the planar continuum robot in the environment. The state consists of the current position(x and y) and the goal position (x_{goal} and y_{goal}) of the continuum robot in the planar coordinate system. The state space is designed to be within the task space of the robot. As the continuum robot moves in the environment, it will have three different curvature values (κ) since it is a three-section robot. Therefore, we use κ for each section as the action space, with each κ limited to the range of $[-1, +1]$.

环境规范是强化学习方法中的关键组成部分。在本研究中，我们直接应用第 2 节中介绍的机器人正运动学和速度运动学，来描述环境中的平面连续体机器人。状态由连续体机器人在平面坐标系中的当前位置 (x 和 y) 及目标位置 (x_{goal} 和 y_{goal}) 组成。状态空间设计在机器人的任务空间内。由于连续体机器人在环境中运动时具有三个不同的曲率值 (κ)，因为它是三节机器人。因此，我们使用 κ 作为每节的动作空间，每个 κ 限制在 $[-1, +1]$ 范围内。

Fig. 5 Schematic diagram of

图 5 示意图



Deep Deterministic Policy Gradient

深度确定性策略梯度 (Deep Deterministic Policy Gradient)

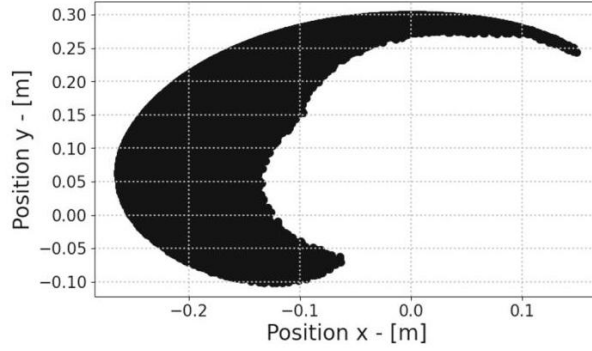


Fig. 7 Task Space of three-section continuum robot

图 7 三节连续体机器人的任务空间

To complete the environment, the reward function must be designed. The proper design of the reward function is crucial for the agent to take meaningful action in reinforcement learning problems. Given that our aim is to control the continuum robot for point-to-point access, it makes sense to create a reward function that calculates the distance to the goal position at each time step. To establish the reward function, we started by computing the Euclidean distance to the goal state utilizing Eq. 17. The resulting distance value was then multiplied by negative one, as demonstrated in Eq. 18. This function multiplies the Euclidean distance between the robot's current state and the goal state by negative one to increase the reward value as the robot gets closer to the target. The reward is negative to discourage the robot from deviating from the goal state, with the penalty increasing as the distance increases. This reward function is the foundation of the RL algorithm for controlling the continuum robot, and shapes its behavior during the learning phase.

为了完善环境，必须设计奖励函数。奖励函数的合理设计对于智能体在强化学习问题中采取有效行动至关重要。鉴于我们的目标是控制连续体机器人实现点对点的到达，设计一个在每个时间步计算到目标位置距离的奖励函数是合理的。为建立奖励函数，我们首先利用公式 17 计算到目标状态的欧氏距离。然后将该距离值乘以负一，如公式 18 所示。该函数通过将机器人当前状态与目标状态之间的欧氏距离乘以负一，使得机器人越接近目标，奖励值越高。奖励为负值，旨在惩罚机器人偏离目标状态，且距离越大惩罚越重。该奖励函数是控制连续体机器人的强化学习算法的基础，塑造了其学习阶段的行为。

$$d_u = \sqrt{(x - x_{\text{goal}})^2 + (y - y_{\text{goal}})^2} \quad (17)$$

$$r = -1 \cdot (d_u)^2 \quad (18)$$

4 Simulations and results

4 仿真与结果

This section discusses the results of the RL algorithm used in simulations. At the start of each simulation episode, the robot is randomly spawned within the task space. The base of the continuum robot is mounted at

point(0,0), and each section of the three-section robot starts from a random position relative to this point. The simulation uses a time step of ΔT equal to 0.1 s. To accomplish the intended task, the robot was trained using the DDPG algorithm with the keras-rl framework [36].

本节讨论了在仿真中使用的强化学习 (RL) 算法的结果。每次仿真开始时, 机器人会随机生成在任务空间内。连续体机器人的基座安装在点 (0,0), 三段机器人中的每一段均从相对于该点的随机位置开始。仿真使用的时间步长 ΔT 等于 0.1 s。为了完成预定任务, 机器人采用 keras-rl 框架 [36] 中的 DDPG 算法进行了训练。

To successfully train the algorithm, at least 5000 episodes are required, with a maximum of 1000 steps per episode to prevent the robot from getting stuck. As a result, the robot needs a maximum of 5,000,000 steps to perform good learning on the targeted region in the workspace. After training the continuum robot on the NVIDIA TESLA P40 Graphical Processing Unit (GPU) accelerator using cudatoolkit version 11.2 and cudnn version 8.1.0 for approximately 14 hours, the 2D position reach task was achieved. To effectively train the neural networks of Actor and Critic, it is important to input appropriate hyperparameters.

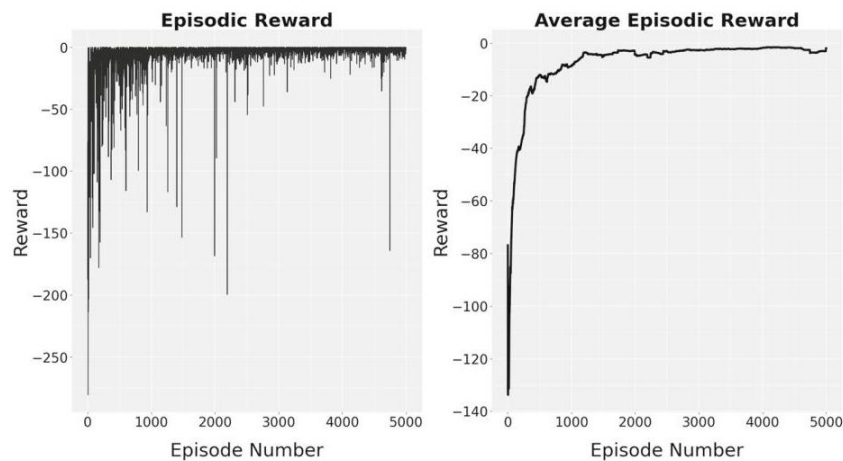
为了成功训练该算法, 至少需要 5000 个回合, 每个回合最多 1000 步, 以防止机器人陷入停滞。因此, 机器人最多需要 5,000,000 步才能在工作空间的目标区域上实现良好的学习。在使用 cudatoolkit 11.2 版本和 cudnn 8.1.0 版本的 NVIDIA TESLA P40 图形处理单元 (GPU) 加速器上训练连续体机器人约 14 小时后, 完成了 2D 位置到达任务。为了有效训练 Actor 和 Critic 神经网络, 输入合适的超参数非常重要。

A 250-episode moving average of the rewards, along with the rewards received for each episode, were saved to evaluate the performance of the RL algorithm (See Fig. 8). The use of a 250-episode moving average facilitates the interpretation of the plots by smoothing out any fluctuations.

保存了 250 回合的奖励移动平均值以及每个回合获得的奖励, 以评估强化学习算法的性能 (见图 8)。采用 250 回合的移动平均有助于通过平滑波动来更好地解读图表。

Fig. 8 Moving average and

图 8 移动平均和



episodic reward per episode

每回合的回合奖励

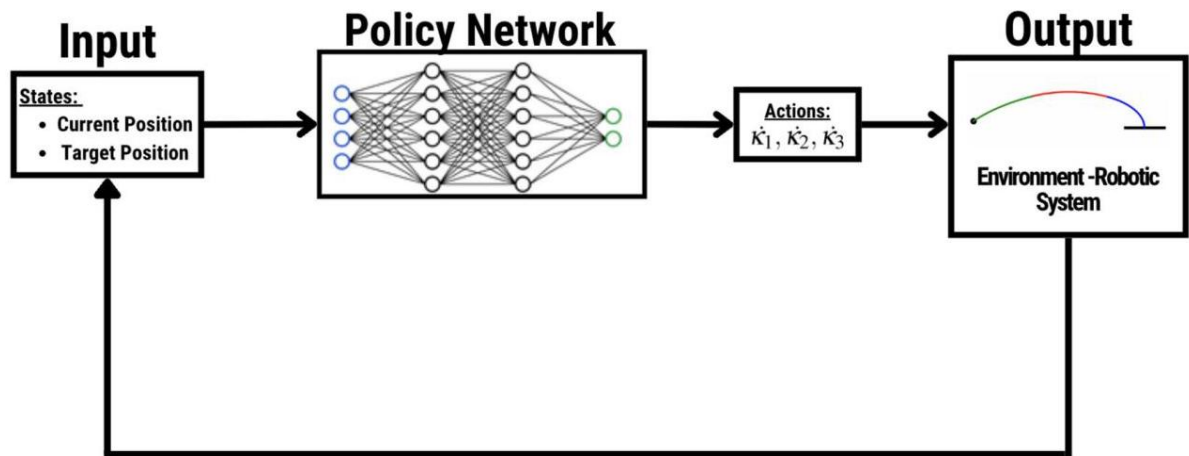


Fig. 9 General control scheme

图 9 通用控制方案

After the training phase was completed, the neural network weights were saved and made ready for use in the control application during the testing phase. The control scheme for the robot is presented in Fig. 9, where an actor network, also known as the policy network, takes the current and goal states as inputs to produce the action $\dot{\kappa}$ for each of the three robot sections at every time step.

训练阶段完成后, 神经网络权重被保存, 并准备在测试阶段的控制应用中使用。机器人的控制方案如图 9 所示, 其中演员网络 (也称为策略网络) 以当前状态和目标状态为输入, 在每个时间步为三段机器人生成动作 $\dot{\kappa}$ 。

Figure 10 depicts some examples of the robot's motion on the X-Y plane during test simulations. To evaluate the performance of the control algorithm, multiple simulations were conducted. Figure 11 (a) and (b) illustrate the total error value and the error values for the X-axis and Y-axis at each step of the continuum robot, respectively. The confidence bands for these figures provide a statistical measure of the consistency of the results across the multiple simulations. The figures demonstrate that the robot is capable of reaching the goal point and maintains its position in subsequent simulation steps. As a result, the error converges to nearly zero after a certain point.

图 10 展示了测试仿真中机器人在 X-Y 平面上的一些运动示例。为了评估控制算法的性能, 进行了多次仿真。图 11(a) 和 (b) 分别展示了连续体机器人每步的总误差值以及 X 轴和 Y 轴的误差值。这些图的置信区间提供了多次仿真结果一致性的统计度量。图中显示机器人能够到达目标点, 并在随后的仿真步骤中保持位置。因此, 误差在某一点后趋近于零。

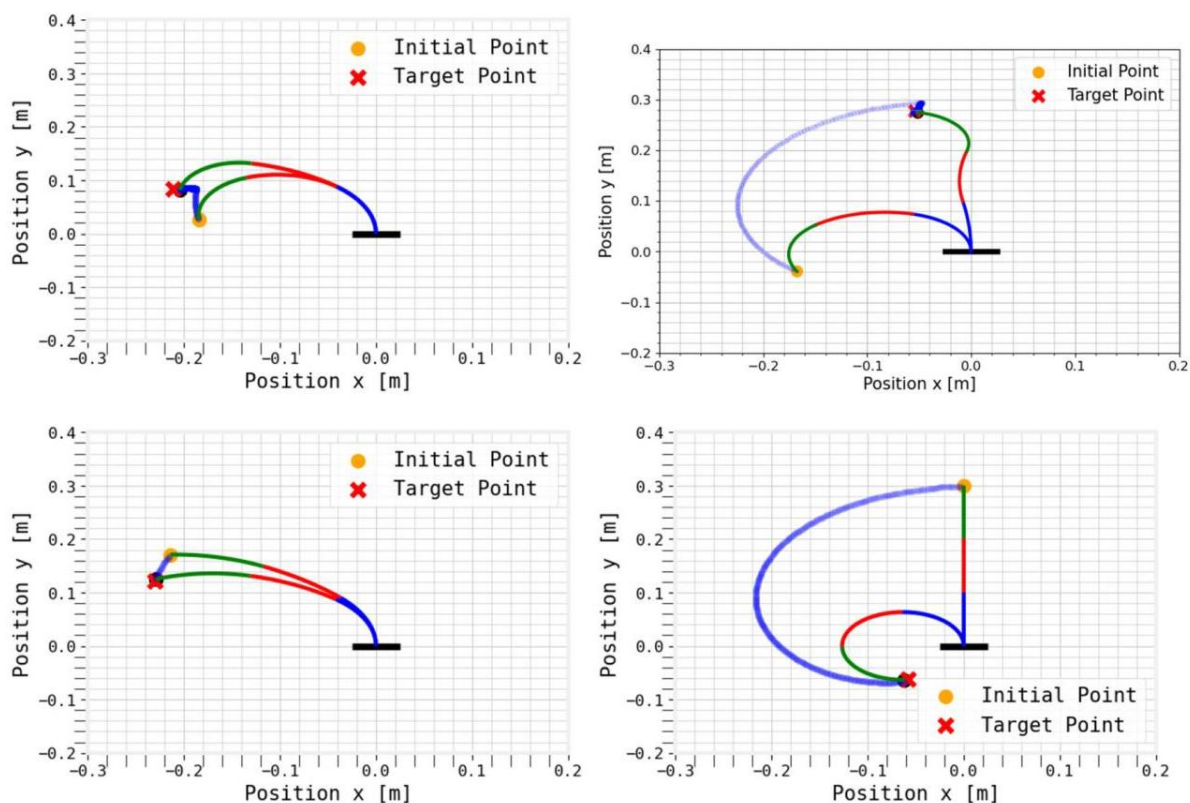


Fig. 10 Frames of a few sample episodes with the robot's trajectory

图 10 机器人轨迹的若干示例回合帧

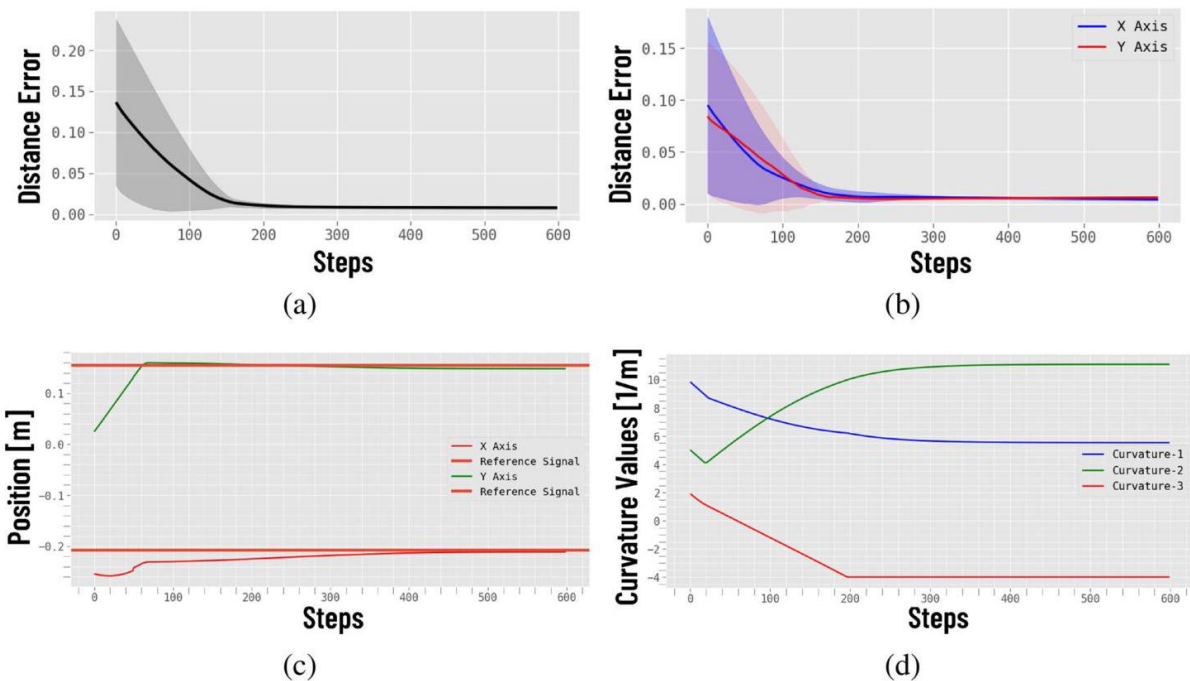


Fig. 11 Total error of the continuum robot (a), error in X and Y axis (b), position in X and Y axis (c), curvature values for each step (d)

图 11 连续体机器人的总误差 (a), X 和 Y 轴的误差 (b), X 和 Y 轴的位置 (c), 每步的曲率值 (d)

Further analysis of the results can be observed in Fig. 11 (c) and (d) that are obtained from single test simulation. Figure 11 (c) shows the position for the X-axis and Y-axis at each step, providing additional insight into the robot's trajectory. Figure 11 (d) displays the curvature value for the three different robot sections at each step during a single simulation. These figures offer a more detailed understanding of the robot's movement and provide evidence of the effectiveness of the control algorithm in achieving the desired results. Overall, the significant point of using RL for a continuum robot is its ability to enable autonomous learning and adaptability in the environment, providing a promising approach for advancing the capabilities and applications of continuum robots.

图 11(c) 和 (d) 展示了单次测试仿真得到的进一步结果分析。图 11(c) 显示了每步 X 轴和 Y 轴的位置, 提供了机器人轨迹的更多细节。图 11(d) 展示了单次仿真中三段机器人每步的曲率值。这些图表更详细地揭示了机器人的运动情况, 并证明了控制算法在实现预期结果方面的有效性。总体而言, 使用强化学习控制连续体机器人的重要意义在于其能够实现自主学习和环境适应性, 为提升连续体机器人的能力和应用提供了有前景的方法。

5 Conclusions

5 结论

Reinforcement Learning (RL) has evolved from being a term in psychology to becoming a common method of learning in computer science. RL is often observed in nature, such as when a cheetah runs at high speeds hours after birth without any instruction. In summary, this paper has demonstrated the application of reinforcement learning in the context of the planar continuum robot control. This study presented a pioneering approach to continuum robot control by melding a specific forward kinematics and velocity kinematics model with an RL control algorithm. The results underscore the efficacy of our method, especially in the context of the planar continuum robot. While our immediate findings are promising, they also pave the way for future research. The novelty of our approach is evident in its adaptability and potential scalability to more complex robotic systems. As the field of robotics continues to evolve, we believe that such integrative methods will play a pivotal role in addressing the multifaceted challenges of robot control. The use of the constant curvature representation for forward kinematics to describe a shape as well as velocity kinematics for the robot's motion allowed for the creation of an environment for RL to learn the motion of the robot from an initial point to a goal point. Traditional reinforcement learning algorithms are not suitable for this problem due to its continuous state and action spaces, which led to the application of the DDPG algorithm for the control of the continuum robot. The paper also discussed the selection of appropriate choices for the reward function, state space, and action space, which are essential components of reinforcement learning systems. Tuning the hyperparameters of the two neural networks used in the DDPG algorithm was also highlighted as necessary to obtain a suitable solution. The results of the simulations show that the DDPG algorithm was able to learn the control of the continuum robot and move it from an initial point to a goal point efficiently. Our future work involves extending our current model to include a spatial movement platform and orientation, which will allow us to capture the full complexity of continuum robot control and make our model more realistic. We will also investigate the use of other reinforcement learning algorithms, like PPO or TRPO, and integrate sensory feedback to enhance the robot's ability to navigate complex environments. Overall, this paper contributes to the growing body of knowledge on reinforcement learning and its application in robotics, particularly in the control of continuum robots. The results demonstrate the potential of RL-based control of continuum robots and highlight

the importance of continued research in this area to achieve significant breakthroughs in robotics technology.

强化学习 (Reinforcement Learning, RL) 已从心理学术语发展为计算机科学中常用的学习方法。强化学习常见于自然界, 例如猎豹出生数小时后无需任何指导便能高速奔跑。总之, 本文展示了强化学习在平面连续体机器人控制中的应用。本研究提出了一种开创性的方法, 将特定的正向运动学和速度运动学模型与强化学习控制算法相结合, 实现了连续体机器人的控制。结果强调了该方法的有效性, 尤其是在平面连续体机器人领域。尽管当前结果令人鼓舞, 但也为未来研究铺平了道路。该方法的新颖性体现在其适应性及向更复杂机器人系统扩展的潜力。随着机器人领域的不断发展, 我们相信此类集成方法将在解决机器人控制的多方面挑战中发挥关键作用。采用恒曲率表示法进行正向运动学描述形状, 以及速度运动学描述机器人运动, 使得构建强化学习环境以学习机器人从初始点到目标点的运动成为可能。传统强化学习算法不适用于该问题, 因为其状态空间和动作空间是连续的, 因此采用了 DDPG 算法来控制连续体机器人。论文还讨论了奖励函数、状态空间和动作空间的合理选择, 这些是强化学习系统的关键组成部分。调节 DDPG 算法中两个神经网络的超参数也被强调为获得合适解决方案的必要步骤。仿真结果表明, DDPG 算法能够学习连续体机器人的控制, 并高效地将其从初始点移动到目标点。未来工作将扩展当前模型, 包含空间运动平台和姿态, 以捕捉连续体机器人控制的全部复杂性, 使模型更具现实意义。我们还将探索其他强化学习算法, 如 PPO 或 TRPO, 并整合传感反馈, 以增强机器人在复杂环境中的导航能力。总体而言, 本文为强化学习及其在机器人学中, 特别是连续体机器人控制中的应用, 贡献了新的知识。结果展示了基于强化学习的连续体机器人控制的潜力, 强调了在该领域持续研究以实现机器人技术重大突破的重要性。

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